De La Salle Holy Cross College

NOVEMBER EXAMINATIONS

2023

Grade 10

MATHEMATICS

Paper 1

Examiner: A. Bux **Time:** 2 hours

Moderators: A. Seewald, J. Blockley and K. Ruperto **Marks:** 100

Name:	Teacher: **AB AS JB KR**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of **20** pages which include **11** questions and a formula sheet on page 2. Please check that your question paper is complete.
2. Read the questions carefully.
3. **Answer ALL the questions on the question paper and hand it in at the end of the examination.**
4. Diagrams are not necessarily drawn to scale.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Ensure that your calculator is set in **DEGREE** mode.
7. Clearly show **ALL** calculations, diagrams, graphs, etc. that you have used in determining your answers. **Answers only will NOT necessarily be awarded full marks**.
8. Round off final answers to **ONE DECIMAL PLACE** unless otherwise stated.
9. It is in your own interest to write legibly and to present your work neatly; anything that is unclear will not be marked.
10. Three blank pages (15, 19 and 20) are included in the paper. If you run out of space for a question, use these pages. Clearly indicate the question number of your answer should you use this extra space.

	Q 1	Q 2	Q 3	Q 4	Q5	Q 6	Q 7	Q 8	Q 9	Q 10	Q 11	Total
Mark												
Total	8	15	8	9	4	12	6	8	12	14	4	100

Remember that God is with you and will help and inspire you in all that you do

GRADE 10 FORMULA SHEET:

$$A = P(1 + in)$$

$$A = P(1 + i)^n$$

$$A = P(1 - in)$$

$$A = P(i - i)^n$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan\theta$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

SECTION A **[44 Marks]**

QUESTION 1 **[8 Marks]**

Solve for x in the following equations, stating any necessary restrictions.

(a) $5^x = \frac{1}{125}$ (2)

(b) $2x(x - 2a) = 0$ (2)

(c) $-4 \leq 3 - x < 11$, represent your answer on a number line. (4)

QUESTION 2 **[15 Marks]**

Simplify fully:

(a) $(-3x^2y^3)^3$ (2)

(b) $\dfrac{2^{3n+2} - 2^{3n}}{2^{3n+2}}$ (4)

(c) $\dfrac{27^{n-4} \times 9^n}{3^n \times 81^{n-3}}$ (4)

(d) $\left(a - 6 + \frac{5}{a}\right) \div \frac{a - 1}{a}$ (5)

QUESTION 3 **[8 Marks]**

Seven years ago, Miss Gibb decided to invest R18 000 in a bank account that paid simple interest at 4,5% p.a. She decides to buy a television set that costs R27 660 now.

(a) Calculate how much interest Miss Gibb earned over the 7 years. (2)

(b) If the average rate of inflation over the last 5 years was 6,7% p.a., calculate the cost of the television set 5 years ago. (3)

(c) At what rate of simple interest should Miss Gibb have invested her money 7 years ago if she intends buying the television set now using only her original investment of R18 000 and the interest earned over the last 7 years? (3)

QUESTION 4 **[9 Marks]**

(a) Matthew is enchanted and writes down the name JULIE over and over again as shown:

JULIEJULIEJULIEJULIE …

What is the 197th letter in his list? (2)

(b) The first four terms of a number pattern are shown below:

8	3	-2	-7

(1) Write down the next two terms of the pattern. (1)

(2) Determine a general formula for the nth term of the pattern. (2)

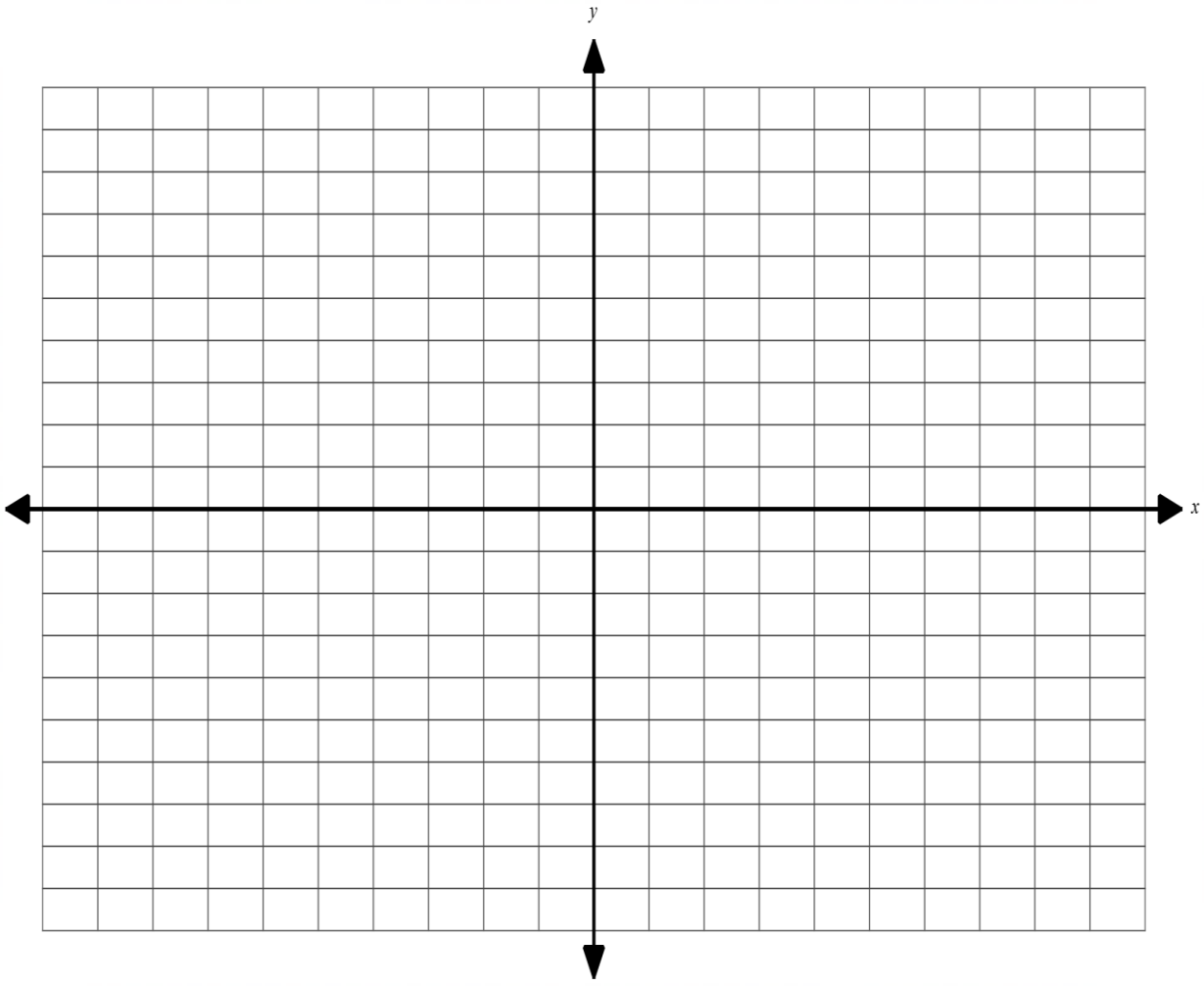
(3) Calculate the value of the thirtieth term. (2)

(4) Which term in the pattern has a value of -492? (2)

QUESTION 5 **[4 Marks]**

Given $h(x) = \frac{3}{x} - 1$.

Sketch *h(x)* on the system of axes below.
Show all intercepts with the axes and asymptotes. (4)



44 marks

SECTION B **[56 Marks]**

QUESTION 6 **[12 Marks]**

(a) Simplify fully:

$$\frac{121^x - 1}{11^x - 1}$$ (2)

(b) Solve for x in the following equation, stating any necessary restrictions.

$$\frac{2x}{x-3} + \frac{5x-3}{9-x^2} = \frac{x}{x+3}$$ (6)

(c) Solve simultaneously for x and y:

$-2x - y = 10$ and $3x - 4y = -4$. (4)

QUESTION 7 **[6 Marks]**

In this question attempts have been made to answer the given questions. Serious errors have been made. Circle the **first error only**. **Correct** only the **line with the error**. You do not need to complete the whole problem.

Attempt at the question	**Correction**
$x^2 = 2x$ $\frac{x.x}{x} = \frac{2x}{x}$ $\therefore \quad x = 2$	
$(x-2)(2x+3) = -6$ $x - 2 = -6 \quad \text{or} \quad 2x + 3 = -6$ $x = -4 \quad \text{or} \quad x = -\frac{9}{2}$	
$2.3^x + 5.3^{x+1}$ $= 3^x(2 + 5 \times 1)$ $= 7.3^x$	

QUESTION 8 **[8 Marks]**

(a) Nonhlanhla invests R12 500. She earns interest at a rate of 9% per annum compounded quarterly. Calculate the effective interest rate **correct to FOUR decimal places**. (3)

(b) Shannen deposits R20 000 into an account at an interest rate of 9% per annum compounded quarterly. Two years later the interest rate changes to 10% per annum compounded annually.
He withdraws R12 000 at the time the interest rate changes.
How much does Shannen have after a total of 8 years?
[Hint: make use of a timeline.]

(5)

QUESTION 9 **[12 Marks]**

(a) In a class of 40 pupils, the following information is true:

- 7 pupils are left-handed
- 18 pupils play hockey
- 4 pupils play hockey and are left-handed
- All 40 pupils are either right-handed or left-handed

(1) Draw a Venn diagram to represent the above information. (4)

(2) How many pupils in the class are right-handed and do not play hockey? (1)

(3) Determine the probability that a pupil chosen at random is:

(i) left-handed or plays hockey. (2)

(ii) right-handed and plays hockey. (2)

(b) One morning Samuel conducted a survey outside De La Salle Holy Cross College to establish how many passengers, excluding the driver, travel in a car. The results are shown in the table below:

Number of passengers, excluding the driver	0	1	2	3	4
Number of cars	7	11	6	5	1

Calculate the probabilty that, excluding the driver, there are more than two passengers in a car. (3)

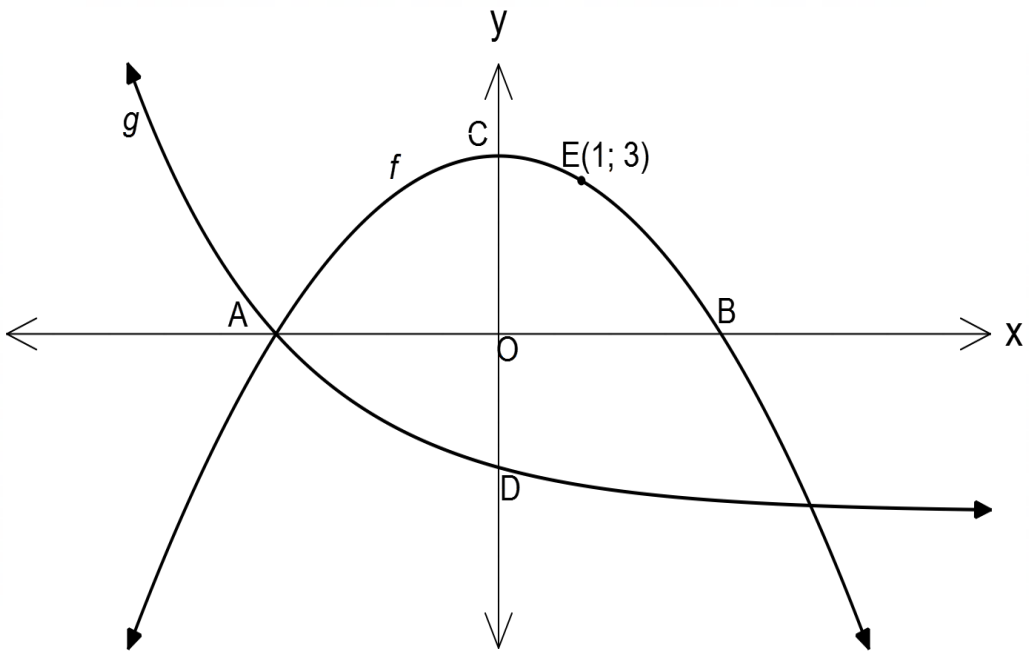
ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL QUESTIONS ARE MARKED.

QUESTION 10 **[14 Marks]**

Sketched below are the graphs of $f(x) = ax^2 + q$ and $g(x) = \left(\frac{1}{2}\right)^x - 4$.

A and B are the *x*-intercepts of *f*. The graphs intersect at A and point E(1; 3) lies on *f*.
C is the turning point of *f* and D is the *y*-intercept of *g*.



(a) Show that the co-ordinates of D are (0; –3). (1)

(b) Show that the co-ordinates of A are (–2; 0). (2)

(c) Calculate the values of *a* and *q*. (4)

(d) Determine the:

(1) length of CD. (2)

(2) equation of a straight line through A and D. (2)

(e) For which value(s) of x is $f(x) > 0$? (1)

(f) Determine the equation of the graph formed when g is reflected about the x-axis. (2)

QUESTION 11 **[4 Marks]**

Ayla was given the following calculation:

$9785627^2 - 9785630 \times 9785624$

Her calculator gives zero as the answer, which is not correct. Use your knowledge of Mathematics to determine the correct answer.
You may not use long multiplication. (4)

56 marks

TOTAL: 100 marks

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